**14** Here is a table of values for $m$ and for $r$

| $m$ | 2 | 6 | 10 | 14 |
|---|---|---|---|---|
| $r$ | 20 | 16 | 12 | 8 |

Harry says,

"$r$ is inversely proportional to $m$ because the values of $r$ decrease by 4 and the values of $m$ increase by 4"

(a) Is Harry correct?
You must give a reason for your answer.

**1 mark**
Show unequal products.

No. $2 \times 20 = 40$, but $6 \times 16 = 96$.

Inverse proportion would make mr constant.

**(1)**

$y$ is inversely proportional to $x^2$

(b) Complete this table of values.

| $x$ | 1 | 2 | 5 | 10 |
|---|---|---|---|---|
| $y$ | 200 | 50 | 8 | 2 |

$y = \frac{k}{x^2}$ $\quad 50 \times 2^2 = k = 200$

**1 mark**
Use $y = k \div x^2$.

**1 mark**
Substitute known values.

At $x = 1$: $y = 200 \div 1^2 = 200$

At $x = 5$: $y = 200 \div 25 = 8$

For $y = 2$: $2x^2 = 200 \rightarrow x^2 = 100 \rightarrow x = 10$

**2 marks total**
All three table values.

**(4)**

**(Total for Question 14 is 5 marks)**

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